Name:

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Date:

Fall 2023 Semester

Instructor:

Samuel Chukwuemeka

Project:

Number Theory Application: Check Digits for International Standard Book Numbers (ISBNs)

Book Title:

Ready Classroom Mathematics

Copyright Year, Edition, and ISBNs:

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Volume 1

ISBN-10: 1495780341

ISBN-13: 9781495780349

Objectives:

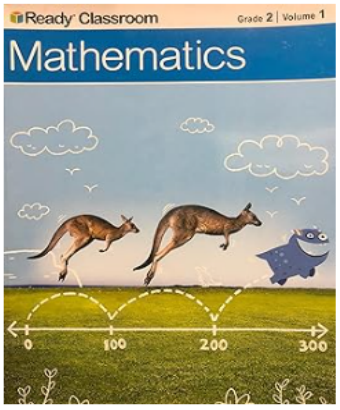
(1.) Verify the check digit for ISBN-10

(2.) Verify the check digit for ISBN-13

(3.) Convert ISBN-10 to ISBN-13

(4.) Covert ISBN-13 to ISBN-10

Calculations:



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1st Objective: Verify the check digits for ISBN-10 (modulo 11)

Given: 1495780341

Approach: Multiplication-Modulo Approach

$1 \times 1 = 1$

$4 \times 2 = 8$

$9 \times 3 = 27$

$5 \times 4 = 20$

$7 \times 5 = 35$

$8 \times 6 = 48$

$0 \times 7 = 0$

$3 \times 8 = 24$

$4 \times 9 = 36$

Sum of the Products

$1 + 8 + 27 + 20 + 35 + 48 + 0 + 24 + 36 = 199$

Sum divided by Modulo

$199 \div 11 = 18\ Remainder\ 1$

$Check\ digit = Remainder = 1$

This is the correct check digit.

<u>2nd Objective: Verify the check digit for ISBN-13 (modulo 10)</u>

Given: 9781495780349

Approach: Multiplication-Modulo-Subtraction Approach

$9 \times 1 = 9$

$7 \times 3 = 21$

$8 \times 1 = 8$

$1 \times 3 = 3$

$4 \times 1 = 4$

$9 \times 3 = 27$

$5 \times 1 = 5$

$7 \times 3 = 21$

$8 \times 1 = 8$

$0 \times 3 = 0$

$3 \times 1 = 3$

$4 \times 3 = 12$

Sum of the Products

$9 + 21 + 8 + 3 + 4 + 27 + 5 + 21 + 8 + 0 + 3 + 12 = 121$

Sum divided by 10

$121 \div 10 = 12\ Remainder\ 1$

$Modulo - Remainder$

$10 - 1 = 9$

This is the correct check digit.

<u>3rd Objective: Convert ISBN-10 to ISBN-13</u>

Given: ISBN-10: 1495780341

To Convert: to ISBN-13

(1.) Remove the check digit, 1: we have 149578034

(2.) Replace it with the check digit for ISBN-13: we have 1495780349

(3.) Append 978 in front of the remaining digits: we have 9781495780349

ISBN-13 is 9781495780349

<u>4th Objective: Convert ISBN-13 to ISBN-10</u>

Given: ISBN-13: 9781495780349

(1.) Remove the check digit, 9: we have 978149578034

(2.) Replace it with the check digit for ISBN-10: we have 9781495780341

(3.) Remove the number, 978, from the front of the remaining digits: we have 1495780341

ISBN-10 is 1495780341

References (APA):

Chukwuemeka, S. (2023). Check digits for International Standard Book Number (ISBN). *Mathematics Education Courses* https://www.mathematicseducation.appspot.com/

Ready Classroom. (2020). *Ready classroom mathematics*. Curriculum Associates